Supplementary Information: Few-shot metal adaptation recovers cross-metal collapse

Hugo Lee Mol-Metal contributors

October 13, 2026

1 Probe-cohort collapse decomposition

The main text reports the collapse on the paper-funnel cohort. On an earlier, smaller probe cohort (rows filtered to ≤ 24 h exposure) the collapse is larger and we additionally measured a same-metal Bemis–Murcko scaffold-split control, which separates the metal axis from the ordinary novel-scaffold penalty.

Table 1: Probe-cohort transfer decomposition: within-metal 5-fold, same-metal scaffold split, and cross-metal LOMO, same scorer. Net metal = scaffold − cross. **MEASURED**: `metrics/by_metric/metal_activity_transfer_seed_stability.json`, `metal_activity_transfer_scaffold.json`.

Metal	Within (5-fold)	Scaffold	Cross (LOMO)	Net metal
Ru	0.8271 [0.8128, 0.8363]	0.7338	0.6179	+0.116
Ir	0.8430 [0.8295, 0.8597]	0.7087	0.5276	+0.181
Rh	0.6974 [0.6429, 0.7381]	0.4690	0.5190	−0.050
Os	0.4960 [0.3133, 0.5867]	0.4900	0.4950	−0.005
Re	0.9480 [0.9083, 0.9819]	0.8465	0.4429	+0.404

A same-metal scaffold split costs 0.10–0.12 on the two metals where the penalty resolves, so roughly half of the raw within-to-cross drop is ordinary novel-scaffold generalisation and the metal axis is the largest single remaining shift. On this smaller cohort Rh ($n = 51$), Os ($n = 36$) and Re ($n = 74$) are too small to gate: a label-permutation null on the fixed cross-metal ranking places all three inside the null (null SD ± 0.09–0.10), and only Ru is significantly above chance after Holm correction. This is the reason the main text adopts the paper-funnel cohort, where every metal is resolvable. **MEASURED**: `metrics/by_metric/metal_activity_transfer_permutation_null.json`, `metal_activity_transfer_significance.json`.

Representation-dependence. Replacing the fingerprint scorer with the chemprop v2 backbone on the probe cohort reproduces the collapse for Ru, Ir and Rh and recovers Re (0.7368, Holm $p = 0.010$), where the fingerprint scores 0.4429. Whether Re transfers is therefore a property of the representation, not of the metal axis. **MEASURED**: `metrics/by_metric/pa3_chemprop_lomo_probe_cohort.json`.

2 Static metal conditioning does not repair the collapse

The main text's repair is adaptation. The obvious alternative — feeding metal identity into a zero-shot model — does not work, and this is the control that motivates the adaptation framing.

Table 2: Static metal conditioning, cross-metal LOMO mean over five metals, paired against the unconditioned baseline. **MEASURED**: `metrics/by_metric/pa15_metalcond_lomo.json`, `l3_metalcond_lomo.json`.

Arm	Cross-metal mean	Paired Δ	Holm p
Unconditioned (baseline)	0.5693	—	—
Metal token	0.5205	−0.0488	1.000
Metal descriptor	0.5480	−0.0215	1.000
FiLM	0.5275	−0.0418	0.984

All three leave the cross-metal mean at or below the unconditioned baseline: the collapse is not a missing-input problem.

3 tmQM pretraining A/B

Table 3: tmQM pretraining A/B on the Ru and Ir activity heads, fixed Bemis–Murcko scaffold split, paired over five seeds; `init` is the only variable. **MEASURED**: `metrics/by_metric/tmqm_pretrain_init_ab_ru.json`, `tmqm_pretrain_init_ab_multimetal.json`.

Metal	Arm	Mean ROC-AUC	Sample std over seeds
Ru ($n_{\text{test}} = 215$)	tmQM init	0.7778	0.0194
Ru	random init	0.7872	0.0201
Ir ($n_{\text{test}} = 56$)	tmQM init	0.7205	0.0399
Ir	random init	0.6719	0.0258

On Ru the paired delta is −0.0094 (t-based 95% CI $[-0.0406, +0.0218]$, $t(4) = -0.84$, 2/5 seeds positive). On Ir the point estimate is +0.0486 but it does not clear the paired t-test (3/5 seeds positive, CI $[-0.0238, +0.1210]$ includes zero, one-sided $p = 0.068$; a 5-seed percentile bootstrap returns $p = 0.012$ but is anti-conservative at that sample size). Across the two metals no Holm-corrected effect is significant, so we make no pretraining claim. The measurement is sound — the paired design cancels seed variance and both arms log the tmQM tensor-load count — the effect is null. An earlier +0.0181 came from a degenerate, molecule-level splitter and does not survive the fix.

4 MoleculeNet backbone control

The control's only job is to show the encoder is a competent organic-property predictor, so that the cross-metal collapse is a property of the axis and not of a backbone that cannot fit.

The D-MPNN wins bbbp and hiv and loses bace and clintox. The two losses exceed seed noise and are carried here rather than in the main text; they are generic ADMET endpoints outside this

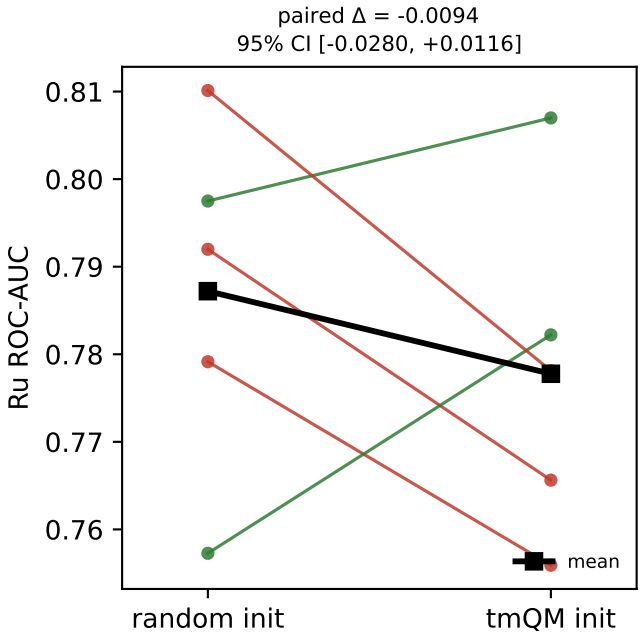


Figure 1: Paired per-seed deltas for the tmQM pretraining A/B on the Ru arm (fixed Bemis–Murcko scaffold split). Two of five seeds move toward tmQM (green), three away (red); the black line is the mean. The interval crosses zero. **MEASURED**: `metrics/by_metric/tmqm_pretrain_init_ab_ru.json`.

paper's target chemistry and do not bear on the metal-axis claims.

5 Tie-rule sensitivity

When duplicate measurements of a ligand disagree, the collapse rule resolves an exact tie toward *active*. This is the anti-conservative direction, so we report the within-metal column under both rules.

The Ir result is robust to the rule; the Ru result is decided by it. The collapse itself is unaffected, being a difference of two columns computed under the same rule.

6 Cohort-construction trap

A preprocessing filter applied to an already-filtered artifact can manufacture a below-chance result. The default probe-cohort snapshot is filtered to short exposure times (≤ 24 h); raising the threshold at training time cannot recover dropped rows, and the loaded cohort stayed byte-identical — the filter silently did nothing. The consequence was a below-chance fold (Rh collapsed to 108 test ligands, AUC 0.3812) with no change to the model. Readers building on MetalCytoToxDB can hit the same trap; the main text's cohort avoids it by constructing from the raw database under the published rule. **MEASURED**: `metrics/by_metric/metal_lomo_auc_wasserstein.json`.

Table 4: D-MPNN and chemprop v2 on MoleculeNet tasks, identical scaffold split, five seeds. "Literature" is the commonly quoted best-of-search constant. **MEASURED**: `metrics/by_metric/moleculenet_auc_dmpnn_5seed_cap512_d4.json`, `moleculenet_auc_chemprop_v2_same_protocol.json`.

Task	This work (D-MPNN 512/4)	chemprop v2 (same protocol)	Δ	Literature
bbbp	0.8994 ± 0.0218	0.8861 ± 0.0135	+0.0133	0.940
hiv	0.7889 ± 0.0189	0.7807 ± 0.0222	+0.0082	0.823
bace	0.7933 ± 0.0250	0.8458 ± 0.0113	−0.0525	0.876
clintox	0.8303 ± 0.0133	0.8847 ± 0.0193	−0.0544	0.894
mean	0.8280	0.8493	−0.0214	0.8832

Table 5: Within-metal AUC under the two tie rules (probe cohort). **MEASURED**: `metrics/by_metric/metal_activity_transfer_tie_sensitivity.json`.

	Ties → active (current)	Ties → inactive (conservative)
Ru	0.8128–0.8363	0.7880–0.8082
Ir	0.8295–0.8597	0.8237–0.8374
Krasnov published: Ru 0.81, Ir 0.73		

7 tmQM bond-perception reference

Table 6: Bond perception on 900 held-out tmQM complexes, identical cohort and per-complex aggregation for every column. **MEASURED**: `metrics/by_metric/tmqm_bond_perception_3seed.json`.

	Learned head	OpenBabel (1)	RDKit (stripped) (2)
bond EXISTENCE F1	$\mathbf{0.9962 \pm 0.0008}$	0.9771	0.9998
exact-type rate	$\mathbf{0.0956 \pm 0.0113}$	0.0300	0.0220
macro-F1(type)	0.4784 ± 0.0055	**0.5051**	0.2809
success rate	1.000	1.000	0.303

The learned head wins two of three axes but loses macro-F1(type) to OpenBabel by 0.0267, consistent across all three seeds.

References

[1] Noel M. O'Boyle, Michael Banck, Craig A. James, Chris Morley, Tobias Vandermeersch, and Geoffrey R. Hutchison. Open Babel: An open chemical toolbox, 2011.

[2] RDKit contributors. RDKit: Open-source cheminformatics. GitHub / conda-forge, 2024. URL `https://github.com/rdkit/rdkit`.

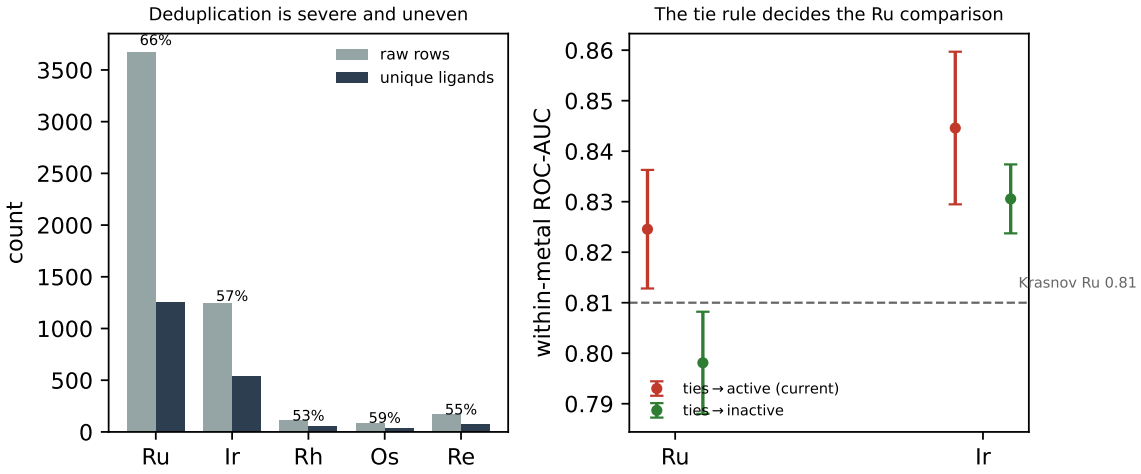


Figure 2: Left: raw rows versus unique ligands per metal (percentage is the duplication rate). Right: within-metal AUC interval across seeds under the two tie rules; the Ru interval crosses the published 0.81 line, the Ir interval does not. **MEASURED**: `metrics/by_metric/metal_activity_transfer_seed_stability.json`, `metal_activity_transfer_tie_sensitivity.json`.